

Peiran Li

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EDUCATION

Dartmouth College, Hanover, NH

M.S. in Computer Science, 50% Tuition Scholarship

Expected Aug 2027

The Chinese University of Hong Kong, Hong Kong

B.Sc. in Computer Science, First Class Honours

Dean's List, Faculty of Engineering (Top 10%); Summer Research Grant (HKD 20,000)

Jul 2025

RESEARCH INTERESTS

Computational imaging and inverse problems; diffusion and generative priors for reconstruction.

PUBLICATIONS

[1] **Peiran Li**, Fei Chen, Xixin Wu. "EEG-based Speech Decoding Based on Multi-mode Joint Modeling." Interspeech 2025.

RESEARCH EXPERIENCE

Single-Shot HDR Imaging with Diffusion Priors

May 2026 – Present

Dartmouth College, Hanover, NH. Research Assistant, supervised by Prof. Adithya Pediredla.

- Formulated the sampling operator as row-wise long/medium/short exposures over a differentiable RAW sensor model, and captured the coded HDR dataset on an ArduCam module.
- Recovered the HDR radiance from that single coded measurement with a pretrained generative prior, optimized self-supervised under an SNR-weighted loss.
- Swapped in posterior-sampling (DPS, DAPS) and MAP/variational (RED-diff, plug-and-play) priors to compare them.

Towards Bandwidth-Limited Imaging with Generative Priors

Aug 2025 – Present

Dartmouth College, Hanover, NH. Research Assistant, supervised by Prof. Adithya Pediredla and Prof. Sotiris Nouisias.

- Sampled 6.25% of pixels with a stratified random mask at native sensor resolution.
- Completed the full image with an SD3 diffusion backbone, conditioning a trained ControlNet on the mask and a nearest-neighbor interpolation of the samples.
- Outperformed downsample-then-super-resolve pipelines (Swin2SR, HAT, StableSR, Stable Diffusion $\times 4$ Upscaler) at matched budget: LPIPS 0.192 vs 0.294, FID 10.99 vs 13.83, and every metric on Urban100.

EEG-based Speech Decoding Based on Multi-mode Joint Modeling

Dec 2024 – Aug 2025

Centre for Perceptual and Interactive Intelligence, Hong Kong. Research Intern, supervised by Prof. Xixin Wu.

- Collected EEG from 11 participants across three speech modes: imagined, intended, and spoken.
- Jointly modelled the three modes rather than training on imagined speech alone, augmenting EEGNet with dynamic masking over temporal-spatial features to reach 34.95% four-way vowel accuracy for imagined speech.

Gaze-Based Weak Supervision for Medical Image Segmentation

Sep 2024 – Nov 2024

The Chinese University of Hong Kong. Research Intern, supervised by Prof. Qi Dou.

- Developed a weak-supervision scheme combining gaze annotation with multi-level learning from discriminative attention; operated eye trackers and preprocessed gaze with CRFs.
- Cut annotation time from 12 hours to 2.2 hours with no loss in segmentation accuracy.

HARDWARE & EXPERIMENTAL SYSTEMS

- Galvo/APD scanning system: built and aligned a galvanometer-mirror scanner with a single-pixel APD, driving the mirrors along resonant Lissajous trajectories to realize sparse sampling optically.
- SPAD laser ranging: time-of-flight depth acquisition with single-photon avalanche diodes, including laser-detector synchronization and photon-timing histograms.
- EEG and eye-tracking acquisition: operated EEG recording systems and eye trackers, and ran the full collection protocol with human participants.